Logic in Computer Science, Prolog 3

BITS Pilani- K K Birla Goa Campus

Total: 8 Marks | Time : 50 Minutes
Weightage: 4% | November 16, 2023
Course No: *CS F214*

Note:

- **Store your file using your ID number. If it is not followed, you will get negative marks. For example if your ID no is 2014A7PS1234G then your file name should be 2014A7PS1234G.pl**
- **Your file should contain three predicates (myack/3, mylisttonum/2, mysudoku/16).**
- **You have to provide definitions of all the auxiliary predicates used and you are not allowed to use any libraries.**
- **There should be no compilation error or warning messages when you run your prolog program. Any error or warning message, your program will not be evaluated and you will get negative marks.**
- **The test cases might have variables as well not only just ground queries. For example, one test case for mylisttonum may be mylisttonum([2,3,4],Y).**

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1. Write a prolog program with the following predicates

   (a) myack(X,Y,Z) returns true iff Z =a(X,Y) where a is the Ackermann's function from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$ is defined by the following equations.

   - a(0,y)= y
   - a(x,0)=a(x-1,1)
   - a(x,y)= a(x-1,a(x,y-1))

   For example, myack(1,2,4) returns true and myack(0,3,5) returns false.
   Query: myack(0,0,X). Answer: X=0
   Query: myack(1,2,X). Answer: X=1
   Query: myack(2,3,X). Answer: X=1
   Query: myack(3,2,X). Answer: X=1

   (2 *Marks*)

   (b) mylisttonum(L,X) returns true iff X is the number represented by the digits in the list L. For example, mylisttonum([2,3,0,1],2301) returns true and mylisttonum([4,3],0) returns false. You are allowed to use in-built arithmetic operators but any other auxiliary predicates has to be defined.
   Query: mylisttonum([1,2,3],X). Answer: X=123
   Query: mylisttonum([0],X). Answer: X=0
   Query: mylisttonum([7],X). Answer: X=7
   Query: mylisttonum([2,3,0,4],X). Answer: X=2304
   Query: mylisttonum([2,5],X). Answer: X=25
   Query: mylisttonum([0,2,3,4],X). Answer: X=234

   (2 *Marks*)

2. Consider 4×4 Sudoku problem: There are sixteen variables, one for each entry in the grid. The domain for each variable is $\{1, 2, 3, 4\}$. The constraints to be satisfied

are the following: for each row, column, and quadrant, the four variables must have distinct values.

Write a prolog program for the predicate mysudoku.
For example, mysudoku(1,4,3,2,3,2,4,1,2,3,1,4,4,1,2,3) returns true. Here the values are entered row by row (which means 4 elements of the first row followed by 4 elements of the second row...). (4 *Marks*)

Query: mysudoku(2,3,1,4,4,1,3,2,R1,R2,R3,R4,R5,R6,R7,R8):
Answer: R1,R2,R3,R4,R5,R6,R7,R8
1,2,4,3,3,4,2,1
3,4,2,1,1,2,4,3
1,4,2,3,3,2,4,1
3,2,4,1,1,4,2,3